AISL-Chapter1

Chapter: 2 - Statistical Learning - Intro Lab

Matrix and Simple Funcs

```
x <- c(1, 3, 2, 5)
x
```

```
## [1] 1 3 2 5
```

```
y = c(1, 4, 3, 8)
y
```

```
## [1] 1 4 3 8
```

```
?c

# ? allows us to view the help func with the function's documentation
```

```
length(x)
```

```
## [1] 4
```

```
length(y)
```

```
## [1] 4
```

```
x + y
```

```
## [1]  2  7  5 13
```

```
ls()
```

```
## [1] "x" "y"
```

```
rm(x, y)
```

```
ls()
```

```
## character(0)
```

```
#rm(lis = ls()) # removes all current objects at once
```

```
?matrix

x <- matrix(data = c(1, 2, 3, 4, 5, 6), nrow = 3, ncol= 3)
```

```
## Warning in matrix(data = c(1, 2, 3, 4, 5, 6), nrow = 3, ncol = 3): data length
## differs from size of matrix: [6 != 3 x 3]
```

```
x
```

```
##      [,1] [,2] [,3]
## [1,]    1    4    1
## [2,]    2    5    2
```

```
## [3,]    3    6    3
```

```
z <-matrix(data = c(1, 2, 3, 4),  2 , 2)
z # other method of creating the matrix
```

```
##      [,1] [,2]
## [1,]    1    3
## [2,]    2    4
```

```
x <-matrix(data = c(1, 2, 3, 4),  2 , 2, byrow = TRUE)
x
```

```
##      [,1] [,2]
## [1,]    1    2
## [2,]    3    4
```

```
x == z
```

```
##       [,1]  [,2]
## [1,]  TRUE FALSE
## [2,] FALSE  TRUE
```

```
sqrt(z)
```

```
##          [,1]     [,2]
## [1,] 1.000000 1.732051
## [2,] 1.414214 2.000000
```

```
z^3
```

```
##      [,1] [,2]
## [1,]    1   27
## [2,]    8   64
```

```
sqrt(z^-1/19087) # you can get creative here and throw whatever at the machine
```

```
##             [,1]        [,2]
## [1,] 0.007238210 0.004178982
## [2,] 0.005118187 0.003619105
```

```
sqrt(z^47*(x/sqrt(87^56))) # see above...
```

```
##              [,1]         [,2]
## [1,] 7.026361e-28 1.620298e-16
## [2,] 1.443763e-20 1.977745e-13
```

```
x <- rnorm(90)
y <- x + rnorm(90, mean = 75, sd = .1)
cor(x,y)
```

```
## [1] 0.9951791
```

```
set.seed(3)
y <- rnorm(90)
mean(y)
```

```
## [1] 0.02449497
```

```
var(y)
```

```
## [1] 0.7605948
```

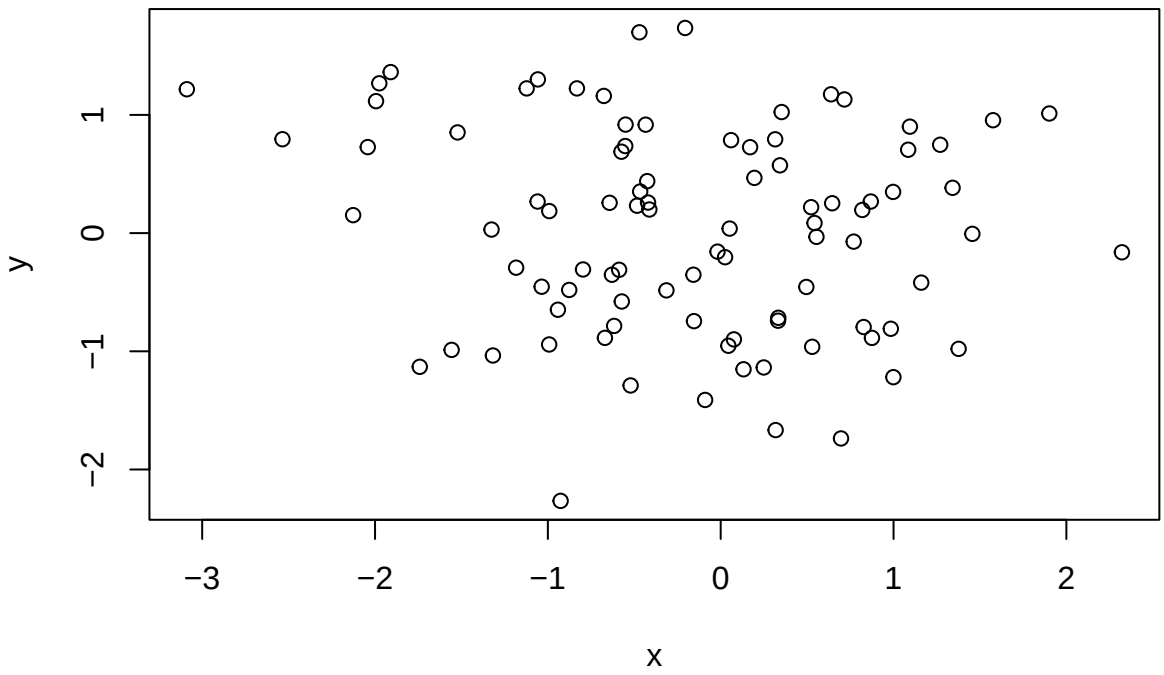
```r
sqrt(var(y)) # sqrt of var = sd
```

```
## [1] 0.8721209
```

```r
sd(y) # same value as above func
```

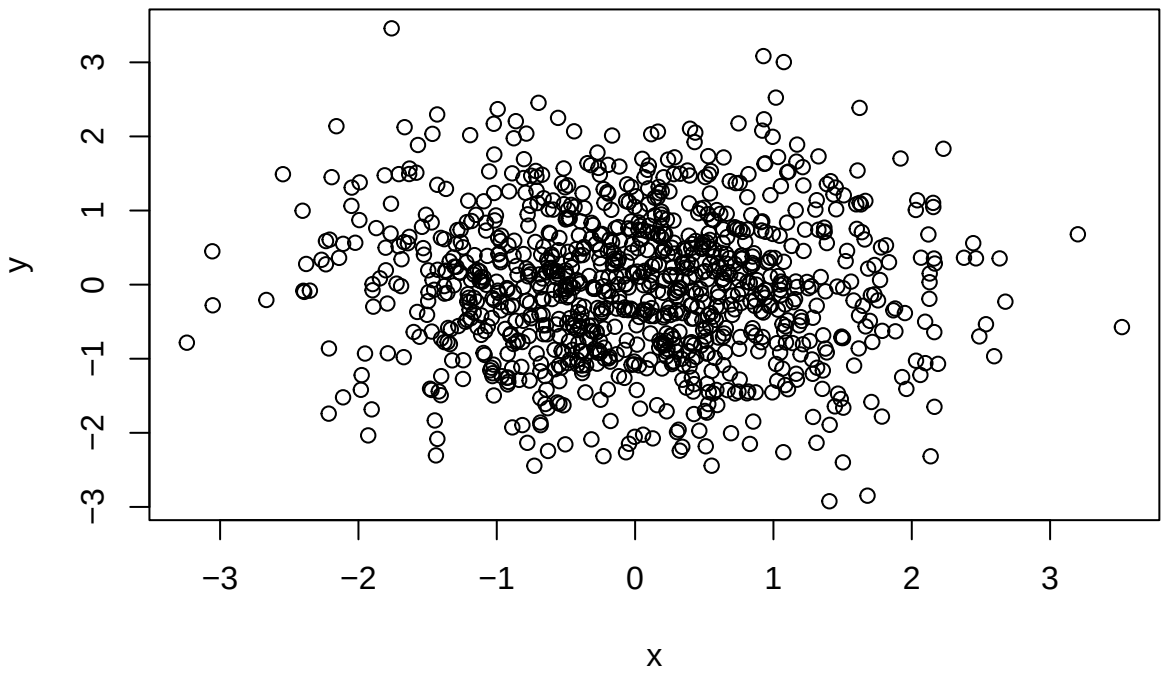
```
## [1] 0.8721209
```

```r
plot(x,y)
```

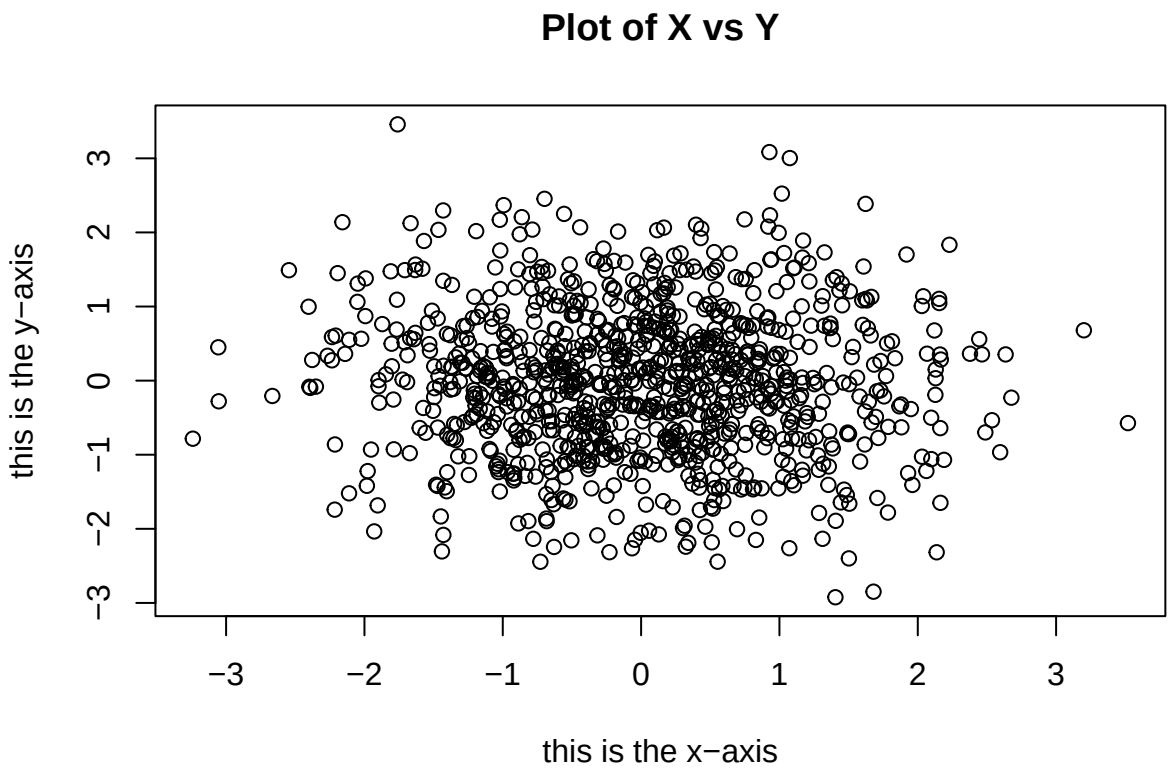


Plotting

```r
x <- rnorm(1000)
y <- rnorm(1000)
plot(x,y)
```




```
plot(x,y, xlab="this is the x-axis", ylab="this is the y-axis", main= "Plot of X vs Y")
```




```
pdf('Figure1.pdf') # We open the pdf here
plot(x,y, col ='green') # Add one graph
x <- rnorm(1000)
y <- rnorm(1000)
plot(x,y) # and another
plot(x,y, xlab="this is the x-axis", ylab="this is the y-axis", main= "Plot of X vs Y") # and another
dev.off() # we close the pdf and save
```

```
## pdf
##   2
```

```
x <- seq(1, 10)
x
```

```
##  [1]  1  2  3  4  5  6  7  8  9 10
```

```
x <- 9:90
x
```

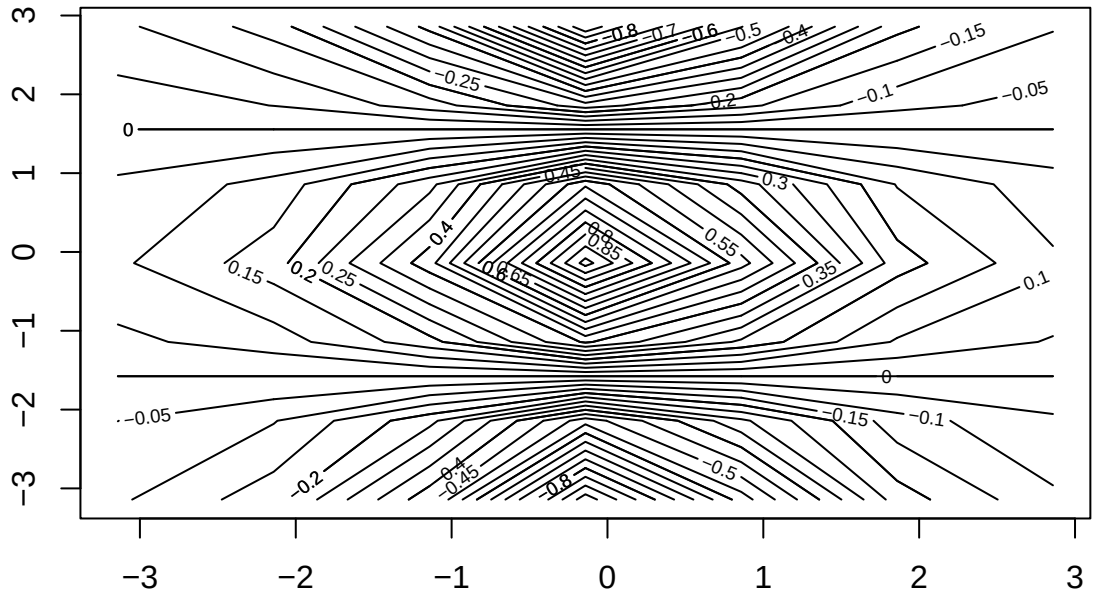
```
##  [1]  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33
## [26] 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58
## [51] 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83
## [76] 84 85 86 87 88 89 90
```

```
x <- seq(pi, -pi, length= 50)
x
```

```
##  [1]  3.14159265  3.01336438  2.88513611  2.75690784  2.62867957  2.50045130
##  [7]  2.37222302  2.24399475  2.11576648  1.98753821  1.85930994  1.73108167
```

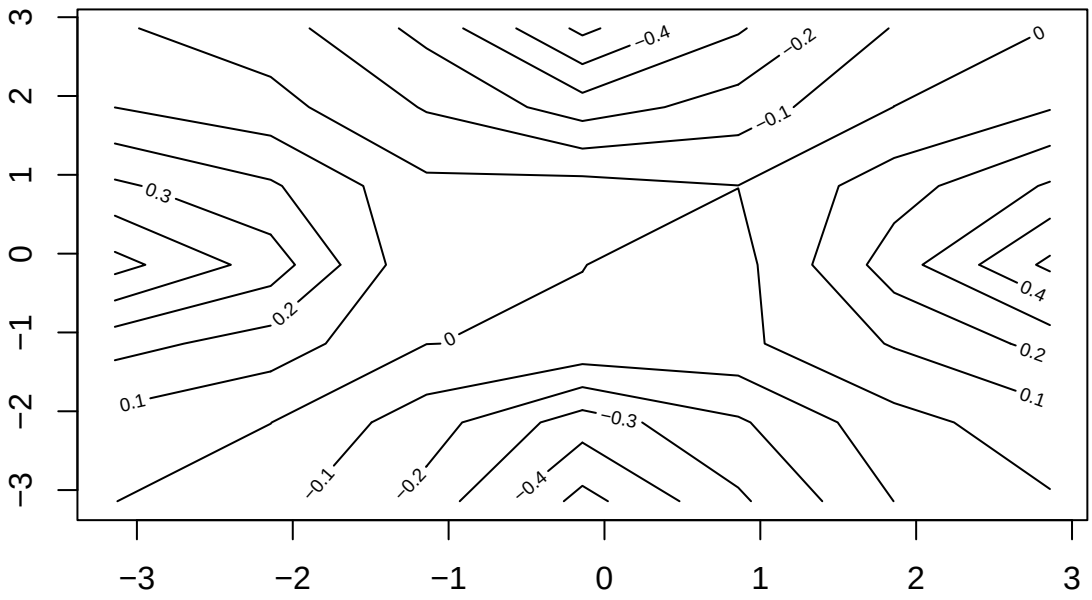
```
## [13]  1.60285339  1.47462512  1.34639685  1.21816858  1.08994031  0.96171204
## [19]  0.83348377  0.70525549  0.57702722  0.44879895  0.32057068  0.19234241
## [25]  0.06411414 -0.06411414 -0.19234241 -0.32057068 -0.44879895 -0.57702722
## [31] -0.70525549 -0.83348377 -0.96171204 -1.08994031 -1.21816858 -1.34639685
## [37] -1.47462512 -1.60285339 -1.73108167 -1.85930994 -1.98753821 -2.11576648
## [43] -2.24399475 -2.37222302 -2.50045130 -2.62867957 -2.75690784 -2.88513611
## [49] -3.01336438 -3.14159265
```

```
x <- y <- seq(-pi,pi)
f <- outer(x, y , function(x, y) cos(y) / ( 1 + x^2))
contour(x, y, f)
contour(x, y, f, nlevels = 45, add = T)
```

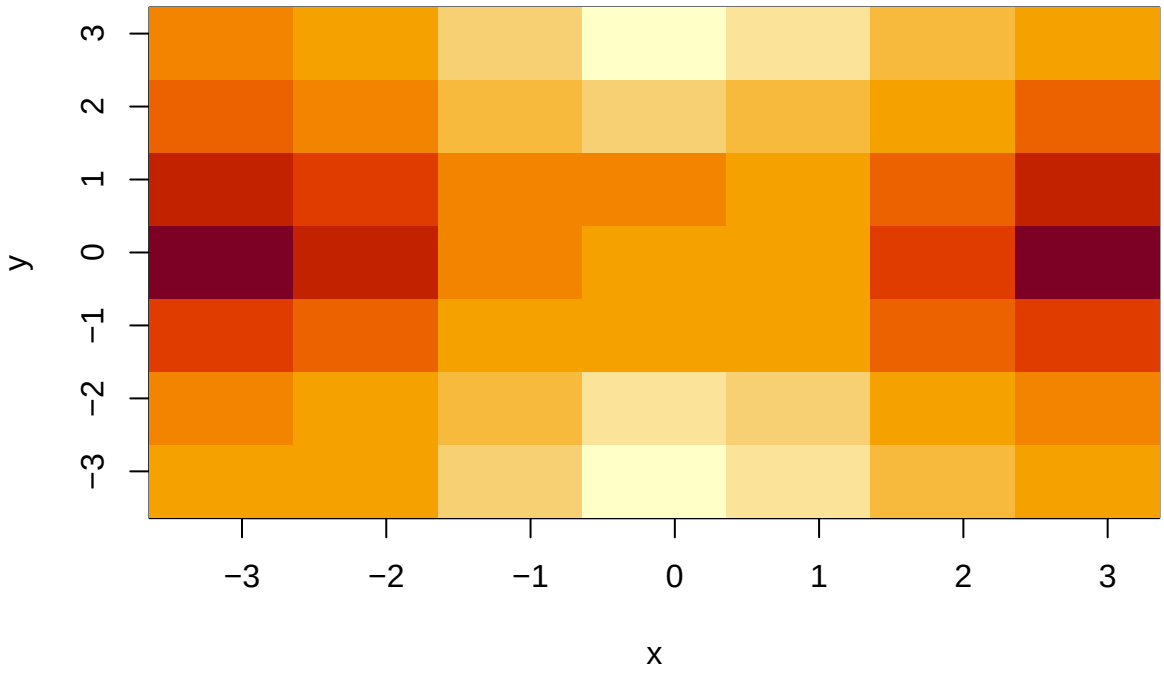


pretty cool looking right?

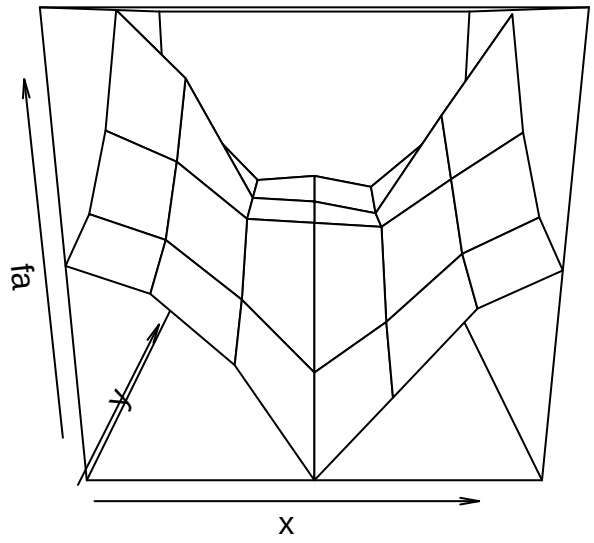
```
fa <-(f - t(f)) / 2
contour(x, y, fa, nlevels= 15)
```



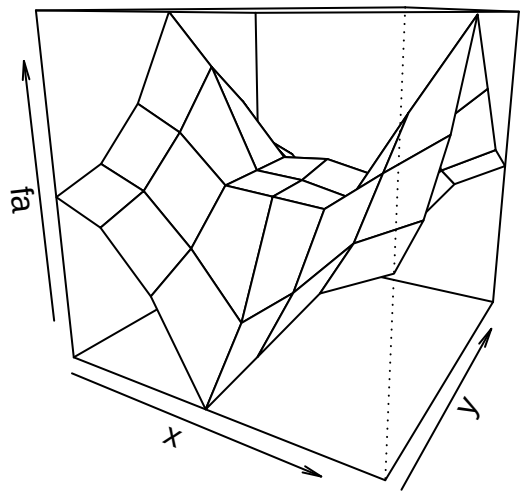

```
image(x, y, fa) # heatmaps
```



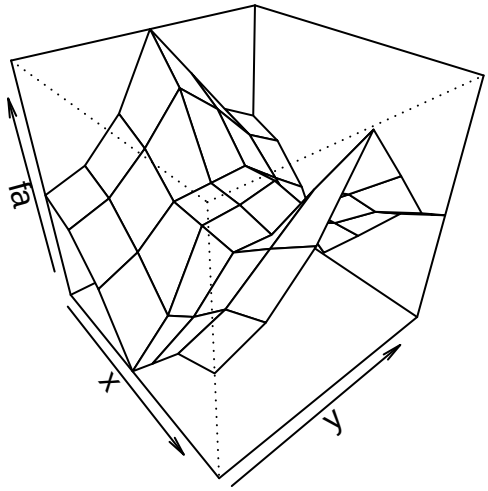

```
persp(x, y, fa)
```




```
persp(x, y, fa, theta = 30)
```




```
persp(x, y, fa, theta = 50, phi = 35)
```



Indexing Data

```
A <- matrix(1:16, 4, 4)
A
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    5    9   13
## [2,]    2    6   10   14
## [3,]    3    7   11   15
## [4,]    4    8   12   16
```

```
A[2,3] # [Row, Column]
```

```
## [1] 10
```

```
A[c(1,3), c(2,4)]
```

```
##      [,1] [,2]
## [1,]    5   13
## [2,]    7   15
```

```
A[1:3, 2:3]
```

```
##      [,1] [,2]
## [1,]    5    9
## [2,]    6   10
## [3,]    7   11
```